

Ikigai -- A Biologically Grounded Digital Organism: Development Report, Day 79

Mura ALife Labs (Formerly Hitoshi AI Labs) -- NeuroSeed Project

Author: Prince Siddhpara, Founder -- Mura ALife Labs (Formerly Hitoshi AI Labs) **Date:** June 22, 2026 **Project:** NeuroSeed **Subject:** Day 79 -- 15 Packs Shipped, 3 Honest Negatives, 0 Reverted. THE COMPOSITIONAL DAY. Pack 285.7 SBC Substrate Flip KILLED (Two Empirical Proofs: Superposition 0%, Redundancy Does Not Fix Noise-Brittleness). Pack 304 + 304.1 + 304.2 Compositional Derive-Not-Store (Store Atoms, DERIVE Relations -- $O(n^2)$ Reach, Self-Taught Atoms From The Oracle, Multi-Relation Chaining Via A DISCOVERED Inverse Rule). Pack 305 + 305.1 RULE DISCOVERY (Organism Mines Its Own Composition Rules By Anti-Unification, Stores Them As HVs In Its Body, SELF-COMPRESSES By Dropping Atoms Rules Make Redundant, Discovers Inverse Schema $conf=1.0$). Pack 306 Fast Path + CRT Factored Decode (Facts $\sim 1.5s \rightarrow <1ms$, $\sim 500x$). Signed-RHC (Math Ring Signed-Complete -- Negatives Via Modular Upper-Half). Pack 300 v0 + 300.1 GENERATION (Speak Grammatical Sentences; LEARN Sentence Templates From Teacher Demos, Not Authored). Pack 301 v0 Coherent Multi-Fact Paragraphs. Pack 302 v0 Reading Comprehension Across Sentences (Made-Up Entity Proof). Pack 287.5 v4 Accent-Folding Parse Fix (espanol/bogota No Longer Fragment To 'ol'). Pack 298 Cache Bit-Compression ($123 \rightarrow 5.5 \text{ B/Entry}$, $1B \text{ Facts} \rightarrow 5.5 \text{ GB}$). Pack 299 Device Profile (1M-Fact Organism = 203 MB, Phone-OK). Pack 303 Real SimpleQA Benchmark ($0\% \rightarrow 2\%$, Honest -- Teacher + Multi-Word-Answer Bottlenecked). Negatives: 285.7, 287.5 v3 (LLM Sycophancy), Free N-Gram (FHRR N-Gram Blur). Bench Holds 30/30 All Day. Zero Hardcoding (Enforced 3x). **Classification:** Research Document -
- Computational Neuroscience / Artificial Life / Cognitive Systems Engineering

1. Objective

Day 78 closed at eight packs shipped: the arithmetic ring fully closed (add+sub+mul+div substrate-native), compositional multi-hop reasoning and a self-built knowledge graph native in `reason()`, active learning wired into the runtime, a $0\% \rightarrow 85\%$ substrate-only continual-learning number on a SimpleQA-style probe, and the SBC binding bug fixed with $O(d)$ constant-time cleanup proven at $N=10k$.

Day 79 opened against the locked eleven-pack queue, led by Pack 285.7 (the SBC substrate full flip) as Prince's stated priority. The day's actual arc, however, was set by two forces: an early empirical falsification that redirected the whole queue, and a single recurring instruction from Prince -- "no hardcoding" -- that was enforced three separate times and shaped the engineering of every capability.

The theme that emerged was **composition**. By close the organism could: store only irreducible atoms and derive relations on demand; teach itself the atoms it lacks from the oracle; discover its own composition rules from its own data; use those discovered rules to chain multi-relation queries; speak its facts as grammatical sentences; learn those sentence templates from teacher demonstrations rather than from authored frames; describe an entity in a coherent paragraph; and read a multi-sentence passage and answer comprehension questions that combine its sentences. The small-device story -- the original motivation behind 285.7 -- closed not through SBC folding but through derive-not-store plus cache compression: a one-million-fact organism now fits in 203 MB.

Day 79 ended at fifteen packs shipped, none reverted, with three honest negatives recorded as first-class results. The substrate-only bench held at **30/30 = 100%** through every landing.

2. Pack 285.7 -- SBC Substrate Flip: KILLED (Premise Falsified)

The locked plan's first priority was to flip the cat-4 substrate from FHRR to Sparse Block Codes (SBC), folding facts into the body to shrink the external cache -- the small-device story. Before wiring anything, two probes tested the load-bearing assumptions.

Probe 1 -- superposition. Bundling N slot-add-bound SBC HVs into one vector and recovering by per-block argmax returned **0% at N = 10..500**, while an FHRR bundle held its classic ~20-item wall (96% at N=50, 60% at N=100). SBC has no analogue of FHRR's conjugate-rotate unbind: superposing facts collides every block (L=10 slots, ~N/10 facts per slot), so argmax recovers noise. The premise "fold many facts into one body HV" is structurally impossible for SBC.

Probe 2 -- noise. SBC per-fact cleanup is fast and exact at zero noise but collapses under additive phasor noise, and the memo's planned rescue -- per-block redundancy and voting -- does **not** fix it: a 9-fold redundant vote at sigma=0.3 scored 2%, and 0% at sigma>=0.5, against FHRR's 100% at all sigma. Sparse codes are fundamentally noise-fragile; FHRR's density is precisely why it is robust. This is a textbook trade-off, not a tunable parameter.

Production queries are noisy by construction (approximate state match, drift). Flipping to SBC would regress recall. SBC's only genuine win -- constant-time cleanup over a large vocabulary at zero noise (~145 us at any V vs FHRR's 9.6 ms at 10k) -- is already covered by the Pack 273 cache and the 285.5 codebook. Prince's call: close 285.7, keep FHRR as the noise-tolerant substrate, and pursue the small-device goal through compositional derive-not-store instead. The negative result is logged so the flip is never retried.

3. Pack 304 -- Compositional Substrate v0 (Derive-Not-Store)

The principle, stated by Prince at Day 78 close: store only irreducible ATOMS (arbitrary facts that cannot be computed, e.g. capital(France)=Paris); DERIVE relations and compositions at query time instead of caching them. This is the math ring (derive any sum, store none) generalised from arithmetic to relations.

The win is combinatorial. From n atomic country attributes the engine answers $O(n^2)$ composite facts -- same-continent comparisons, attribute inheritance, multi-hop chains -- with zero cache entries for the derived answers. A new module `ikigai/cognition/compositional.py` `CompositionEngine` provides `atom(rel, entity)` (read-only cache lookup via canonical question templates) and `derive(text)` (structural relation-algebra rules: inheritance `continent of capital == continent`, same-attribute comparison, numeric comparison, attribute chains). Honesty is preserved by keeping the rules as grammar (relation algebra), not facts -- no country or continent is named in the module; every concrete value comes from a cache atom.

It was wired native in `general_reasoner.reason()` as `do_derive=True` (default), dispatched after arithmetic and before multi-hop, so composites are DERIVED rather than served from storage. The migration was the real demonstration: 21 `continent(capital)` composites (Pack 293 bootstrap, two phrasings = 42 cache entries) were dropped and replaced by 21 fundamental `continent(country)` atoms; the cache went 1613 -> 1592 while answerable composites went up. Gate: continent-of-capital 10/10, same-continent 210/210, multi-hop 10/10 all via `method=derive`, ZERO cache growth across 226 derivations, 30/30 intact, 10.8x derive:store ratio. A `__delitem__`/pop was added to `CompactAnchorCache` for the drop. Saved; backup recorded.

4. Pack 304.1 -- Self-Taught Atoms (Doctrine-Clean)

Pack 304's continent bootstrap used a curated value dict -- the drift Prince flagged. 304.1 corrected it to the LLM-as-data-oracle doctrine: the organism supplies the QUESTIONS (country entities), the TEACHER supplies the atom VALUES through Pack 294 active learning, with no curated value dictionaries.

The organism self-taught currency, language, and population atoms for self-enumerated countries. The `compositional.py` comparison rules were generalised to any relation via a `_norm_rel` alias map, numeric comparison was added (`_to_num` parses "67 million" / "1.4 billion"), and a `_claim()` mechanism was introduced so that a matched comparison template with a missing atom returns ('unknown', 'derive') rather than falling through to active learning -- a compositional comparison is not an atomic fact to teach, and letting the teacher absorb it would pollute the cache. Gate: same-currency 40/40, same-language 40/40, currency-of-capital chain 3/3, 86 composites derived with zero cache growth, 30/30. Saved.

5. Pack 305 -- Rule Discovery v0

Where 304 applied authored composition rules, 305 made the organism MINE its own. Built on the existing `schema_inducer.py` (Plotkin least-general-generalisation anti-unification), a new `rule_discovery.py` `RuleMiner` searches a hypothesis space of rule schemas -- inheritance `r_out(link(x)) == r_tgt(x)`, synonymy `r_a(x) == r_b(x)` -- over the organism's atom store, promoting rules by support and confidence. This is inductive logic programming: we supply the rule SHAPES, the organism discovers which concrete rules hold. Inventing new shapes is out of scope and stated as the honest boundary.

CompositionEngine gained an enumerable AtomStore (triples/entities/relations, self-populating on every atom() hit -- no curated lists), a learned_rules store persisted via a _comp_state entry added to integrate.py's _PERSIST_ATTRS, an atom_del for self-compression, and a require_learned gate. The killer demonstration is self-compression: the miner discovered continent o capital (support=12, conf=0.75) from co-present atoms; with require_learned=True the discovered rule drove derivation 24/24, while clearing the rules dropped it to 0/24 -- proving the discovery is load-bearing, not the authored regex; then the 12 now-redundant continent(capital) atoms were dropped (1640 -> 1628) with recall intact. The organism learned a regularity and shrank its own memory. Rule and atom index persist across reload (verified). Saved.

6. Pack 305.1 -- Rule Discovery v1 (Substrate-Native HV Rules + Inverse Schema)

Three additions. **Substrate-native HV rules:** each learned rule lives in the body as a bound phasor HV (role(type) (x) role(field0) (x) role(field1), blake2b-seeded unit phasors, FHRR Hadamard bind); rebuild_rule_bank/rule_match do cosine cleanup; sanctioned_inheritance routes through the HV bank. Round-trip verified at sim=1.000; HV-sanctioned derivation 12/12; emptying the HV bank dropped it to 0/12 -- the HV rules are load-bearing. **Native autonomous discovery:** IkigaiOrganism.discover_rules() mines from the organism's own atom index with link-versus-attribute auto-classified from store structure (classify_relations: a relation is a LINK if its values are themselves subjects) -- no external lists; hookable into the sleep/idle tick. **Richer schema:** mine_inverse discovered the round-trip rule inverse capital o country at conf=1.00 (a genuinely new shape), and the miner correctly REJECTED bad candidates at conf=0.00.

Prince enforced "no hardcoding" a second time here: the gate's ground truth was de-hardcoded -- math truth computed in Python, capital regression by a before/after stability snapshot rather than a literal dict. A real persistence bug was also fixed: _load_state had been reloading triple keys as character-tuples (tuple("a\tb")), corrupting the atom index on every reload. End: learned_rules=3, HV bank=3, all persist. Saved.

7. Pack 306 -- Fast Path + CRT Factored Decode (Facts < 1 ms)

Prince's demand: "facts can be done in less than 50 ms." Profiling showed reason() was running the expensive cat-4 recall_action (142s cold rebuild / 77ms hot + a 9.6k-vocabulary codebook matmul) on every call, even when a cheap path would answer. Two fixes. A **fast path** (_fast_answer) resolves arithmetic, derivation, multi-hop, and exact-cache hits before the heavy machinery, returning early; the heavy ICL/active-learn path runs only when nothing cheap and confident resolves. And in residue_hdc.py, a **CRT factored decode**

(`add_factored/sub_factored/decode_factored` via `per-modulus _residue_of + CRT, O(Kd)`) *replaced the* *O(ranged)* brute cosine decode that cost ~300 ms per arithmetic call. Result: cache hit 0.06 ms, arithmetic 0.69 ms, derivation 0.03 ms, full 30/30 in 10 ms total (0.3 ms/query) -- the 50 ms target beaten by ~150x. Math remained exact.

8. Signed-RHC -- Math Ring Signed-Complete

Pack 304.2 (below) surfaced a real limitation: RHC subtraction returned positive CRT residues, so $a-b < 0$ decoded wrong. The fix is substrate-native: the modular ring natively represents $-k$ as $\text{prod}-k$, so `decode_factored(signed=True)` maps the upper half ($x > \text{prod}/2$) to its negative. `math_eval` decodes subtraction signed; `add/mul/div` stay unsigned. Gate: $3-12 = -9$, $5-20 = -15$, $7-8 = -1$, positives unchanged, ring intact. The arithmetic ring is now `add + sub(signed) + mul + div`, all substrate-native, all complete.

9. Pack 304.2 -- Generic Multi-Relation Chaining (Via A Discovered Rule)

The compositional thread's deepening: `derive` now handles "what is the of the of " = `rel1(rel2(X))` for any relation pair (`_CHAIN2`). The inner hop resolves directly or via a LEARNED inverse rule -- new `reverse_atom` (scan the atom index) and `sanctioned_inverse` read the inverse rules discovered in 305.1. "Currency of the country of berlin" resolves `country(berlin)` by reverse-lookup of the capital inverse to germany, then `currency(germany)` to euro. Gate: 9/9 chained derivations (buenos aires -> argentina -> argentine peso, lima -> peru -> sol, stockholm -> sweden -> swedish krona), all `method=derive`, recall and math intact. The pack that uses the rule the organism discovered itself one pack earlier -- the day's compounding made explicit.

10. Pack 300 v0 -- Schema-Framed Generation (Speak Sentences)

The organism answered in single words. The existing free-LM GenerationEngine, tested first, was degenerate on the current language substrate ("the capital of france is" -> "so would then so so if..."). So v0 took the reliable path (Prince: "both -- schema v0 now, free-LM next"): a new `sentence_frame.py` `SentenceFramer` transforms the query structure into a declarative and slots the substrate-derived answer -- grammatical by construction. `IkigaiOrganism.say(query)` handles capital, continent-of-capital, yes/no membership, same-attribute comparison, and arithmetic. Gate: 11/11 grammatical sentences across query types. "What is the capital of france" now answers "The capital of france is paris."; "is the capital of japan in europe" answers "No, the capital of japan is not in europe."

11. Pack 300.1 -- Language Template Discovery (Learn How To Speak)

Prince's idea -- "tell the teacher to teach LANGUAGE" -- is the no-hardcoding upgrade of 300 v0, which authored its sentence frames. The organism now LEARNS sentence templates by induction from teacher demonstrations, the same move as 304 -> 305. A new `language_teach.py` `LanguageTeacher` (on the existing `schema_inducer`) takes teacher fact-to-sentence demonstrations, replaces the slot values with named placeholders, and anti-unifies across demonstrations to a learned template -- e.g. `{VAL}` is the capital of `{SUBJ}` at specificity 1.00 -- then applies it to any fact. `say()` uses the learned template first, the schema framer as fallback.

Prince enforced "no hardcoding" a third time, sharply, when an initial version added a hardcoded tail-filler list and verb list to clean output; both were reverted, and grammaticality is now judged by the LEARNED template's specificity (the teacher's structure is the grammar authority), with zero English word lists. Templates live in `_lang_templates` (persisted), never `b_self`; `b_self` was verified pristine (1652 -> 1652). Gate: capital and currency templates learned and generalised 8/8 to held-out entities; a junk-entity demonstration was correctly rejected. Saved.

12. Pack 301 v0 -- Coherent Multi-Fact Speech

`IkigaiOrganism.describe(entity)` gathers every fact the organism knows about an entity (probing its known relation grammar over the cache), speaks each via learned template or schema frame, and assembles a topic-anchored paragraph. Gate: eight entities produced ≥ 3 -sentence descriptions, all on-topic, every fact correct. Real output for Germany: "Berlin is the capital of germany. The continent of germany is europe. Germany uses the euro as its. The language of germany is german." Visible data flaws ("as its" tail, "ol", accent-stripped "bogot") were honestly attributed to teacher quality, not the mechanism, and motivated the data-quality work below.

13. Pack 287.5 v3 (Negative) + v4 -- Data Quality

The visible junk in 301 prompted a verify-before-absorb attempt. v3 first tried hardcoded badword/length filters (Prince flagged it), then pivoted to a doctrine-clean teacher self-confirmation (ask the oracle yes/no on its own answer). This FAILED: LLM sycophancy confirmed wrong facts 6/6. Self-confirmation does not discriminate. Reverted; logged as a negative with the lesson that the real fix is parsing, not verification.

v4 found the actual root cause: `_parse_answer`'s ASCII word regex was FRAGMENTING accented words -- "espa(n-tilde)ol" split to `['espa', 'ol']`, and the last-word parser returned 'ol'. The fix is a general Unicode NFKD `_ascii_fold` applied before the regex (n-tilde -> n, a-acute -> a), so "espanol"/"bogota"/"munchen" stay whole. Safe (parse-path only,

existing cache hashes unchanged). Unit 10/10. The data-quality problem was a parse bug, not a verification problem.

14. Pack 298 + 299 -- Cache Compression And The Device Profile

298: `cat4_packed_cache.py` `PackedAnchorCache` replaces the Python dict (123 B/entry of real per-object overhead) with sorted int64 keys + auto-width value-ids + an interned value table, bisect lookup. Measured at 100k: 10.0 B/entry in RAM (12.3x), and 5.5 B/entry when delta-encoded and zlib-compressed (22.5x) -- in the 3-10 B target. A 1B-fact cache: 123 GB -> 5.5 GB stored. Exact recall parity. Minimal-perfect-hash to drop keys entirely is the noted path to ~3 B.

299: the device profile. The substrate body is constant in fact count (191 MB), the packed cache grows at ~12 B/entry. A one-million-fact organism totals **203 MB** -- comfortably inside a 512 MB phone budget; ten million totals 311 MB. The small-device story 285.7 set out to tell is told instead by derive-not-store (keeps the fact count low) plus 298 compression plus this profile.

15. Pack 302 v0 -- Reading Comprehension

`IkigaiOrganism.read_passage(text)` parses a multi-sentence passage into atoms (sentence-to-fact, the inverse of the 300.1 templates, with the relation explicit in the text) and absorbs them, so comprehension questions that combine sentences answer via derive. Proved with a made-up entity so success is comprehension, not recall: after reading "The capital of Zogland is Zar. The continent of Zogland is Mu.", the organism answered "what continent is the capital of Zogland in" -> "mu" via derive, combining two just-read sentences; the question returned garbage before reading. The reading twin of generation: read -> atoms -> derive-comprehend.

16. Pack 303 -- Real SimpleQA Benchmark (Honest)

The official OpenAI SimpleQA dataset (4326 questions) was downloaded and the Pack 296 continual-learning pipeline run on a 60-question sample. Phase 0 cold substrate-only: 0/60. Phase 1 one self-teach pass: 47 facts active-learned. Phase 2 post-teach substrate-only: 1/60 = 2%. The honest result -- 0% -> 2% on REAL SimpleQA, against 296's 0% -> 85% on easy capitals -- is diagnostic. Two bottlenecks: the 8B R1-distill teacher's accuracy is low on obscure SimpleQA facts, and recall returns noise-floor tokens rather than the taught value for multi-word gold answers ("Dr. Kalam", "May

16, 2015") behind long questions. The continual-learning mechanism works; real-SimpleQA performance is bottlenecked by teacher capacity and multi-word-answer storage, both real and both fixable. A publishable baseline, reported without inflation.

17. Free N-Gram Fluency -- Negative (The Final Chase)

The day's last effort, at Prince's request, was to make free novel-recombination generation cohere. The first attempt absorbed a teacher corpus into the language bank, but the corpus was R1 chain-of-thought ramble, and generation was byte-identical (the junk-saturated cooccur bank dominated). The second attempt fixed both -- self-distilled a clean corpus from the organism's own templated sentences and absorbed into a fresh clean lang bank. This escaped the "so/if/would" attractor (real domain words appeared: "the capital of sweden is europe...") but the n-gram walk still collapsed on function words ("the the of is", unique-ratio 0.55, valid-bigram 44%). The deeper finding: FHRR n-gram recall over a modest bank is blurry -- high-frequency function words dominate `combined_ngram_candidates`, and they cannot be down-weighted without a hardcoded stopword list (against the standing rule). Free fluency is a real multi-pack arc -- it needs sharper decoding (resonator cleanup at the generation step, or the frame-bigram FSM actually locked), far more clean data, and higher-order-gram weighting. Schema-framed and learned-template generation remain the shipped, working fluency path. Negative recorded.

18. Close -- Day 79 Metrics

Fifteen packs shipped, three honest negatives, none reverted after shipping.

Shipped: 304, 304.1, 304.2 (derive-not-store, self-taught atoms, multi-relation chaining); 305, 305.1 (rule discovery, HV-stored, self-compressing, inverse schema); 306 (fast path + CRT decode, facts <1ms); signed-RHC (math ring signed-complete); 300 v0, 300.1 (schema-framed and learned-template generation); 301 v0 (coherent paragraphs); 302 v0 (reading comprehension); 287.5 v4 (accent parse fix); 298 (cache 22.5x); 299 (device profile). Pack 303 ran as an honest real-SimpleQA baseline.

Negatives: 285.7 (SBC flip, two empirical falsifications), 287.5 v3 (LLM sycophancy defeats self-confirmation), free n-gram (FHRR n-gram blur).

The day's signature was compounding: 305.1 discovered the inverse rule; 304.2 used it to chain `currency(country(berlin))`; that chain surfaced the negative-math gap; signed-RHC closed it -- all in one session. And the small-device goal that opened the day under a falsified premise (285.7) closed under a sound one: a one-million-fact organism in 203 MB, its cache compressible 22.5x, its facts answered in under a millisecond.

Production state: organism.ikg 191 MB; cache 1652 atoms; learned_rules 3 (HV-stored, persisted); learned language templates 2; substrate-only bench 30/30 = 100% held through every landing; zero hardcoding on the production path, enforced three times by Prince. Every save backed up.

Pending into Day 80: multi-word answer storage and a frontier teacher (to lift 303); the free-fluency arc (resonator decode + locked grammar FSM + scaled clean corpus); junk-entity hygiene via rules-validating-data; the minimal-perfect-hash cache floor; Pack 278 v2 legacy deletion (deferred as high-regression-risk cleanup); Pack 295 git-LFS reproducible state (blocked on a git-init decision). One standing outbound obligation: the Kanerva bump on 2026-07-01 if no reply.

Trigger keyword for post-compact memory reload: DAY 79 COMPOSITIONAL.